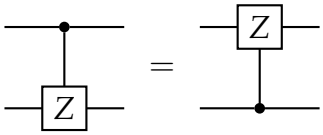
Exercises 12

Exercise 1 (CNOT gymnastics)**.** *Prove the following circuit identities:*

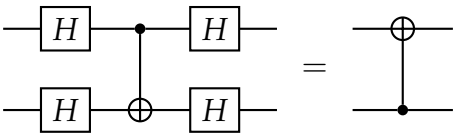
1. *Switching X and Z:*



2. *Controlled-Z is out of control:*

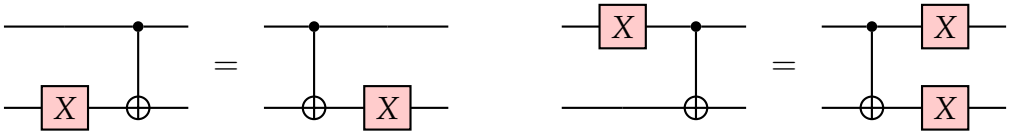


3. *Whaaaat?*

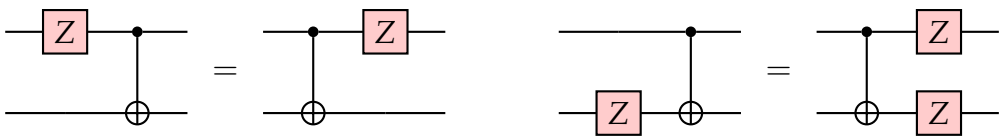


Exercise 2 (Propagation of Pauli errors)**.** *Prove the following circuit identities:*

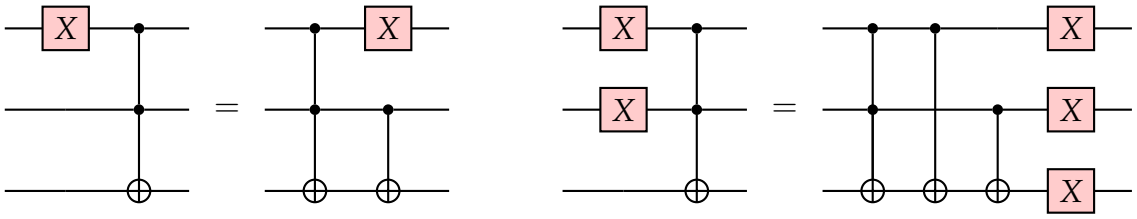
1. *Pauli X moves through* CNOT*:*



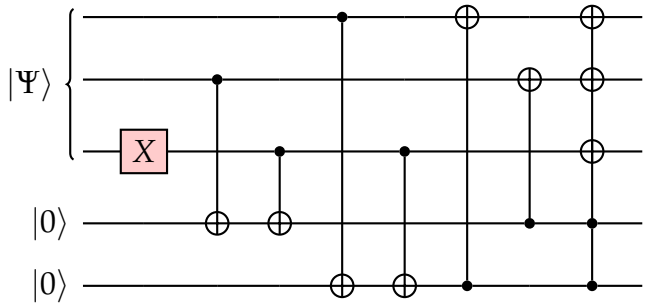
2. *Pauli Z moves through* CNOT*:*



3. *Pauli X moves through Toffoli:*



4. *Assume an X error occurs on the last qubit of the state $|\Psi\rangle$. Propagate this error through the circuit that implements the detection and correction steps of the three-bit repetition code. What effect does this error have once it is moved to the right-hand side of the circuit?*



Exercise 3 (Measuring non-commuting observables)**.** *Let $|\psi\rangle = |0\rangle$. Consider the following two scenarios that correspond to measuring the observables Z and X in different order:*

1. *First measure $|\psi\rangle$ in the standard basis, then in the $|\pm\rangle$ basis (this is the same as first measuring the Z observable and then the X observable).*

2. *First measure $|\psi\rangle$ in the $|\pm\rangle$ basis, then in the standard basis (this is the same as first measuring the X observable and then the Z observable).*

Denote the measurement outcomes in the two cases by $a_1, b_1 \in \{0, 1\}$ and $a_2, b_2 \in \{0, 1\}$, respectively. What are the probability distributions of the strings $a_1 b_1$ and $a_2 b_2$?

Exercise 4 (Commuting matrices can be simultaneously diagonalized)**.** *Let A and B be two normal (and hence diagonalizable) matrices. For simplicity, assume they are 2×2 and invertible. Show that if $AB = BA$ then there is a unitary matrix U such that $UAU^\dagger$ and $UBU^\dagger$ are both diagonal.*

Exercise 5 (Stabilizers of Bell states)**.** *Recall that the four Bell states are as follows:*

$$|\beta_{00}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle), \qquad |\beta_{01}\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle),$$
$$|\beta_{10}\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle), \qquad |\beta_{11}\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle).$$

Note that $|\beta_{00}\rangle$ is stabilized by X_1X_2 and Z_1Z_2:

$$(X \otimes X)|\beta_{00}\rangle = |\beta_{00}\rangle, \qquad (Z \otimes Z)|\beta_{00}\rangle = |\beta_{00}\rangle.$$

Find the stabilizers of the other Bell states.

Exercise 6 (`Cirq`: Steane code)**.**